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Digital product passport - Data exchange protocols

Passeport numérique des produits - Protocoles d'échange de données

Digitaler Produktpass - Protokolle zum Datenaustausch

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European foreword

This document (EN 18216:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital Product Passport - Framework and System”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For the relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

Introduction

0.1 Background

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains - supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of digital product passports will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

To support the implementation of DPPs, 8 standards have been developed so far:

— EN 18219:2026 – Digital product passport – Unique identifiers

— EN 18220:2026 – Digital product passport – Data carriers

— EN 18216:2026 – Digital product passport – Data exchange protocols (this document)

— EN 18222:2026 – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

— EN 18223:2026 – Digital Product Passport – System interoperability

— EN 18221:2026 – Digital product passport – data storage, archiving, and data persistence

— EN 18239:—[1] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246:—[2] – Digital Product Passport – Data authentication, reliability and integrity

0.2 Overview

A digital product passport (DPP) is a dynamic digital record that contains information about a product throughout its lifecycle. For DPPs to be effective and universally accessible, standardized data exchange protocols and frameworks need to be in place. Standardization and harmonization of these protocols ensure that actors of the DPP - such as manufacturers, suppliers, retailers, consumers, repairers, waste treatment facilities, and regulatory authorities - can access, extract, utilize, and update the shared product passport information seamlessly depending on actors' information needs and authorization. The subsequent sections of this document outline the standardization for data exchange protocols and data formats.

In addition, this document includes informative annexes to support practical understanding and implementation:

— Annex A: Data exchange protocol requirements;

— Annex B: Data exchange;

— Annex C: Secure communication;

— Annex D: Examples of systems adaptable to data exchange protocols.

[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

1 Scope

This document defines a standard for secure and efficient data exchange protocols and data formats to be used for the digital product passport. Data exchange protocols establish the rules and procedures that systems follow when communicating and exchanging information. Data formats define the structure and presentation of that information so it can be understood and processed correctly by the involved systems. Together, protocols and formats ensure that data can be exchanged in a manner that is secure, interoperable, reliable, and compatible across various platforms and sectors.

This will guarantee that data are human and machine-readable, structured, searchable, and transferable through an open, interoperable network without vendor lock-in.

a) Secure communication:

   This document defines protocols that ensure secure and authenticated data exchange between systems, ensuring that data are protected against unauthorised access and, when necessary, only authorized entities can access the information.

b) Interoperability for data exchange:

   The protocols and data formats defined in this document support easy integration with existing data exchange systems, ensure compatibility of protocols and formats across various sectors and supporting a wide range of applications and use cases.

c) Ease of use and integration:

   Ensure that the identified protocols and formats can be implemented easily, especially for mobile devices, and are user-friendly in order to facilitate widespread adoption.

d) Data integrity:

   The protocols and data formats defined in this document ensure the integrity of information linked to physical objects and electronic data throughout the entire value chain, extending to the product's or asset's end-of-life.

e) Documentation and discoverability:

   The protocols and formats are available to individuals without specialized knowledge, enabling broader adoption across sectors.

In order to promote interoperability, reduce costs for businesses, and align with existing European regulations and initiatives, this document considers the data exchange protocols and data formats already in use in other legislations. Relevant existing standards are integrated into the development process to ensure consistency and coherence with industry practices and regulatory frameworks.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 21778:2017, *Information technology — The JSON data interchange syntax*

EN 301549:2021, *Accessibility requirements for ICT products and services*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
identifier
digital identifier
sequence of characters associated with digital, non-digital, or abstract entities, such as books, images, reports, metadata records or events

[SOURCE: ISO 24619 [1], 3.2.1]

3.2
data exchange
storing, accessing, transferring, and archiving of data

[SOURCE: ISO 15531-43 [2], 3.1.5]

3.3
identification
process of recognizing an object in a particular domain as distinct from other objects

[SOURCE: EN ISO/IEC 24760-1 [3], 3.2.1]

3.4
authentication
verification that a claimed identity is correct

[SOURCE: ISO/IEC/IEEE 8802-1AR [4], 3.2]

3.5
data integrity
property that data has not been altered or destroyed in an unauthorised manner

Note 1 to entry: In the context of secure communication, data integrity ensures that data transmitted between parties remains unaltered and intact from the moment it leaves the sender to the moment it reaches the receiver. This means that the data has not been tampered with, modified, or corrupted during transmission – whether accidentally or through malicious actions.

[SOURCE: ISO 7498-2:1989, 3.3.21, modified - note to entry added [5]]

3.6
secure communication
mechanism of transmitting data between systems in a way that ensures its confidentiality, integrity and authenticity

3.7
digital product passport
DPP
digital record of product characteristics throughout its lifecycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability

3.8
controlled DPP data
information on digital product passport whose access is controlled based on the user's access rights

Note 1 to entry: User: person who interacts with a system, product or service [SOURCE: ISO 26800:2011, 2.10; modified, Notes changed]

4 Data exchange protocols for technical interoperability

The data exchange protocols specified in this clause shall be used for standardized DPP access.

The data exchange protocols shall maintain confidentiality and integrity of DPP data.

HTTP over TLS (HTTPS)

Protocol: HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP [6], used for secure communication over a computer network.

Standards:

— TLS (Transport Layer Security): TLS 1.2 (e.g. defined by [7]) shall be the minimum version supported. The use of TLS 1.3 (e.g. defined by [8]) or later is strongly recommended. Use of older versions (TLS 1.0, 1.1, and all SSL versions) is prohibited.

— HTTP/2 shall be the minimum version supported. The use of HTTP/3 [9] is recommended. Older versions than HTTP/2 shall not be used.

The API specification based on the data exchange protocol should follow the architectural style of RESTful.

EN 18222 [14] specifies the HTTPS interaction with the API.

5 Data formats

The data format listed below shall be used for syntactic interoperability.

a) JSON (JavaScript Object Notation, ISO/IEC 21778:2017) is a human and machine)readable data-interchange format used to transmit data between a server and a client.

In addition to the abovementioned, any of the following message formats may be used based on http content negotiation [10]:

b) XML (eXtensible Markup Language) [11] is a markup language and file format for storing, transmitting and reconstructing arbitrary data. It defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.

c) JSON-LD (JavaScript Object Notation for Linked Data) [12] is a human-readable data format that provides context and links data, enhancing interoperability and integration between different data sources. Its representation shall be processable by a regular JSON parser, allowing the possibility of including linked data context for advanced semantic processing.

For human readable representation, rendering, or delivery of DPP content shall conform to the relevant accessibility requirements of EN 301549:2021.

The following shall be provided based on HTTP content negotiation:

The content of the DPP shall be provided and rendered according to W3C HTML standards, and should be tested across a range of browser technologies and platforms.

d) HTML (Hypertext Markup Language) is the standard markup language for documents designed to be displayed in a web browser. It defines the content and structure of web content and is often assisted by technologies such as Cascading Style Sheets (CSS) and JavaScript.

Internal storage of DPP data is handled in EN 18221 [13].

Annex A
(informative)

Data exchange protocol requirements

A.1 General introduction to data exchange protocols

A data exchange protocol is a set of rules and standards that govern how data are transmitted, received, and interpreted between different systems or organizations. In the context of digital product passports (DPPs), these protocols are essential for enabling seamless communication and interoperability among various stakeholders, such as manufacturers, suppliers, retailers, consumers, and regulatory authorities.

Data exchange protocols ensure that the information contained within a digital product passport – such as product specifications, origin, materials, compliance certifications, and sustainability metrics – is consistently formatted and securely transmitted. This consistency allows different software applications, platforms, and systems to understand and utilize the data effectively, regardless of the underlying technologies they employ.

Annex B
(informative)

Data exchange

B.1 Security and access control

The data exchange protocols like the ones mentioned in Clause 4 fundamentally enable secure communication between systems over the internet. These protocols should impose requirements to ensure data security, encryption, and overall integrity during data communication.

B.2 Ease of use and integration

The clause provides indication on how to implement identified protocols including the case of mobile devices.

Selecting the right data exchange standards is not just about technical specifications; it is also about how these standards fit into the organisations' workflow and how easily they can be adopted by different users and between actors. This section provides an overview of the data exchange standards, focusing on their ease of use and integration for consumers, small and medium-sized enterprises (SMEs), and larger businesses. Table B.1 summarizes the key points for each standard, highlighting how they impact and how they can be utilized by different user groups.

Table B.1 — Overview of ease of use and implementation of data exchange protocols

Standard	Consumers	SMEs	Large Businesses/ Organisations
HTTPS (HTTP over TLS)	**Ease of use:** Consumers interact with HTTPS daily when browsing websites or using online services, often without realizing it. Modern web browsers automatically handle HTTPS connections.	**Ease of use:** Setting up HTTPS for a website or application is made easy by services of third-party providers, which provide SSL/TLS certificates.	**Ease of use:** Larger businesses typically have IT departments or IT vendors familiar with HTTPS implementation.
	Integration: Not applicable	**Integration:** SMEs can integrate HTTPS by installing certificates on their web servers.	**Integration:** can integrate HTTPS by installing certificates on their web servers.
RESTful APIs	**Ease of Use:** Consumers use applications that rely on RESTful APIs without needing technical knowledge.	**Ease of use:** Developer-friendly. Several frameworks and tools that simplify creating and consuming RESTful APIs are available.	**Ease of use:** Developer-friendly. Several frameworks and tools that simplify creating and consuming RESTful APIs are available.

Standard	Consumers	SMEs	Large Businesses/ Organisations
	Integration: Not applicable	**Integration:** SMEs can easily integrate RESTful APIs into their products and services, enabling them to communicate with other applications and services over the internet.	**Integration:** Larger business and organisations already have knowledge and experience with APIs and probably already offer services based on APIs

B.3 Data integrity

B.3.1 General

Data integrity refers to the accuracy, consistency, and trustworthiness of data throughout its entire lifecycle. In the context of secure communication, it ensures that data transmitted between parties remains unaltered and intact from the moment it leaves the sender to the moment it reaches the receiver. This means that the data has not been tampered with, modified, or corrupted during transmission—whether accidentally or through malicious actions. This section provides a description of how data integrity is maintained when using the data exchange protocols:

— HTTP over TLS

— RESTful APIs

The following clauses explain how the various data exchange protocols preserve data integrity.

B.3.2 HTTP over TLS

a) Message Authentication Codes (MACs):

   1) Integrity checks: HTTPS uses the Transport Layer Security (TLS) protocol, which incorporates MACs to ensure data integrity. MACs are cryptographic hash functions applied to the message data and a shared secret key.

   2) Verification: When data are received, the MAC is recalculated and compared with the transmitted MAC. If they match, the data are considered intact.

b) TLS handshake and cipher suites:

   1) Secure negotiation: during the TLS handshake, the client and server agree on a Cipher Suite that includes a specific hash function (e.g. SHA-256) for integrity checks.

   2) Sequence numbers: TLS uses sequence numbers to prevent replay attacks and ensure that messages are processed in the correct order.

c) Certificate validation:

   1) Trusted authorities: certificates issued by trusted certificate authorities (CAs) help ensure that the entities involved are legitimate, preventing man-in-the-middle attacks that could compromise data integrity.

B.3.3 RESTful APIs

Underlying HTTPS protocol: inherited security: RESTful APIs often operate over HTTPS, leveraging TLS's data integrity mechanisms like message authentication codes.

a) Message validation:

   1) Input validation: servers validate incoming data to ensure it meets expected formats and values, preventing malformed or malicious data from causing unintended effects.

   2) Response validation: clients can validate responses using checksums or hash comparisons when provided.

b) Digital signatures and tokens:

   1) JSON web tokens (JWT): Tokens can include signatures (using HMAC (hash-based message authentication) or RSA (Rivest–Shamir–Adleman algorithms) to verify the integrity of the token data.

   2) OAuth 2.0 and OpenID Connect: tokens used for authentication and authorization can be validated to ensure they haven't been tampered with.

c) Idempotent operations:

   1) HTTP methods: use of safe and idempotent HTTPs methods (like GET, PUT) helps maintain integrity by ensuring consistent outcomes.

Annex C
(informative)

Secure communication

C.1 General

Secure communication defines the mechanism of transmitting data between systems in a way that ensures its confidentiality, integrity and authenticity. This means that the data exchanged are protected from unauthorised access, tampering, or interception by malicious actors. Therefore, this subclause will outline how the data exchange protocols meet the requirements for secure communication of the following technologies:

— HTTP over TLS

— RESTful APIs

C.2 Role of HTTPS and RESTful APIs in secure communication

C.2.1 HTTPS (using TLS 1.2 or 1.3)

HTTPS ensures secure, encrypted communication between the user's mobile app, desktop software, or embedded system and the DPP service. This prevents man-in-the-middle attacks or eavesdropping during data transmission. The TLS protocol secures communication by using asymmetric public key infrastructure. This type of security system uses two different keys to encrypt communication between parties:

a) The private key – this key is controlled by the owner of a website and kept private. This key resides on a web server and is used to decrypt information encrypted by the public key.

b) The public key – this key is available to everyone to securely interact with the server. Information encrypted by the public key can only be decrypted using the private key.

C.2.2 RESTful API*s*

RESTful APIs use the underlying HTTPS protocol to enable secure, stateless data exchange between the client and server. The security of RESTful APIs primarily relies on the HTTPS protocol, which ensures encryption and secure data transmission. RESTful APIs ensure secure communication by:

a) Statelessness: Each request to the server is independent and includes all the information needed for processing. This ensures that sensitive data, such as authentication tokens, is always transmitted over HTTPS, maintaining encryption during transfer.

b) Security: RESTful APIs implement appropriate measures to ensure confidentiality, integrity, availability and authenticity.

c) Rate Limiting: To prevent abuse, REST APIs implement: rate limiting per client, provide rate limit notification, support rate limit exemptions for authorized services, and log rate limit events.

d) DDoS Protection: REST APIs implement appropriate traffic filtering, support traffic rate limiting, provide attack protection, implement mitigation measures, and maintain service availability.

e) Error Handling and Status Codes: RESTful APIs use HTTP status codes to communicate the outcome of a request to the client, ensuring transparency in how the API operates. Errors related to authentication (e.g. 401 unauthorised) or authorization (e.g. 403 Forbidden) help indicate failed security checks, protecting against unauthorised access.

Annex D
(informative)

Examples of systems adaptable to data exchange protocols

The requirements on data exchange protocols can be adapted to the following systems.

This list is non-exhaustive. Other solutions are possible.

a) AS4 Profile of ebMS 3.0. The OASIS ebMS 3.0 Standard combined multiple Web Service standards to create a single comprehensive specification for defining the secure and reliable exchange of documents using Web Services. (AS4 Profile of ebMS 3.0 Version 1.0 (oasis-open.org))

b) Asset Administration Shell: The Asset Administration Shell (AAS) is the digital representation of an asset. The AAS consists of a number of submodels in which all the information and functionalities of a given asset – including its features, characteristics, properties, statuses, parameters, measurement data and capabilities – can be described. It allows for the use of different communication channels and applications and serves as the link between objects and the connected, digital and distributed world. The AAS is designed to be flexible and interoperable allowing it to operate over various data exchange protocols suitable for industrial environments.

c) EDI (Electronic Data Interchange): It is the computer-to-computer exchange of business documents in a standardized electronic format between trading partners. EDI relies on various data transfer protocols to transmit documents.

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of (EU) Regulation 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request *M/604 Amd 1* to provide one voluntary means of conforming to essential requirements of (EU) Regulation 2024/1781 of 13 June 2024 of the European Parliament and of the Council of the European Union establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC.

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative Clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Regulation and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements of (EU) Regulation 2024/1781	Clause(s)/subclause(s) of this EN	Remarks/Notes
Article 10.1.d	4, 5	
Article 11.a	4, 5	the presumption of conformity only concerns the technical exchange protocol and data formats
Article 11.h	4	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products and services falling within the scope of this standard.